

Alexander Yao

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Education

University of Maryland

College Park, MD

Bachelor of Science in Computer Science, Minor in Applied Mathematics

Aug 2023 - Expected Dec 2026

- **Relevant Coursework:** Algorithms, Advanced Data Structures, Compilers, Parallel Computing, Computer Systems, Computer and Network Security, Programming Languages, Web Development, Probability and Statistics

Experience

Computer Network Research Assistant

May 2025 - Feb 2026

George Mason University

Fairfax, VA

- Engineered a Python AsyncIO and QUIC media-delivery system for 20 Mbps real-time video streams, achieving sub-150 ms end-to-end latency in simulated network conditions.
- Designed congestion-control and packet-scheduling logic that reduced packet loss by 50% and improved stream stability by 30% across 20+ simulation runs.
- Built bidirectional Python and Android client-server components over QUIC and instrumented distributed nodes to measure latency, throughput, packet loss, and resource usage.

Cybersecurity Research Intern

Jun 2024 - Aug 2024

Zhejiang University

Hangzhou, China

- Generated interprocedural control-flow graphs and performed taint analysis on Android app APIs, reducing per-app audit time by 93% from 30 minutes to under 2 minutes using Soot and FlowDroid
- Identified 50+ privacy policy violations by comparing data-flow graphs against legal disclosures; presented findings at an international security research conference with 200+ participants.

Projects

NBA Referee Analytics | *React, Vite, FastAPI, Python, Docker, Nginx*

Jun 2026

- Built a full-stack analytics application with a React/Vite frontend and FastAPI backend to explore NBA referee trends from publicly available statistics.
- Implemented API endpoints and service-layer calculations for season and split filtering, foul differential, a weighted home-bias index, z-score outliers, and cross-season consistency analysis.
- Containerized the frontend and backend with Docker Compose and Nginx reverse proxying; delivered interactive visualizations for referee comparisons and derived metrics.

Racket Compiler | *Racket, C, x86-64 Assembly*

Jan 2025 - May 2025

- Built a multi-stage compiler from Racket to x86-64 assembly, including tail-call optimization for deep recursion with constant stack usage.
- Designed a 64-bit calling convention and dynamic tagging system for runtime type safety, arity validation, and low-overhead value representation.
- Implemented a greedy register allocator to reduce stack spills and a C runtime using setjmp and longjmp for hardware-level exceptions and non-local control flow.

Video Performance Tracker | *Python, PostgreSQL, Docker, AWS*

Dec 2025

- Built a Python ETL pipeline with SQLAlchemy to ingest and normalize YouTube and TikTok engagement metrics into PostgreSQL on AWS RDS.
- Developed a Playwright-based TikTok collection service with session management and proxy rotation; calculated normalized engagement rates for cross-platform comparisons.
- Containerized the application with Docker Compose and deployed it to AWS EC2 using persistent volumes and environment-specific configuration.

Technical Skills

Languages: Python, Java, C, Rust, OCaml, JavaScript, TypeScript, SQL, Racket, x86-64 Assembly

Frameworks & Libraries: React, Vite, FastAPI, Node.js, AsyncIO, Pandas, Playwright, SQLAlchemy, Soot

Infrastructure & Tools: Docker, Docker Compose, Nginx, AWS EC2, GitHub Actions, Linux, Git, PostgreSQL

Concepts: CI/CD, Algorithms, Data Structures, Compilers, Static Analysis, Taint Analysis, QUIC/UDP, Congestion Control, Parallel Computing